USE CASE SPECIFICATION · UC-PROF-01 / CURRENT BACKEND

Publish and Maintain Appointment Hours

University Resource Allocation and Scheduling System

Primary actor: Professor (Instructor)	Other actors: Students requesting meetings; appointment service

Goal

Publish bounded weekly meeting hours so students can request eligible appointments without seeing private calendar details.

Precondition

The professor has an active Instructor account and a configured university time zone.

Main success scenario

1. Open Office Hours and choose a weekday and start/end time.
2. Enter explicit valid-from and valid-through dates; optionally exclude individual dates.
3. Submit the publication. The server validates ownership and the time pattern.
4. Review current published patterns; remove a pattern when it is no longer applicable.
5. Students receive only feasible appointment slots after the system excludes known conflicts and personal blocks.

Alternative outcomes and constraints

- Invalid dates, times or pattern: reject without publishing.
- Removing a pattern changes eligibility; existing appointments retain their own lifecycle and conflict checks.
- Another person cannot edit or remove this professor’s patterns.

Result

Appointment-hour patterns are saved or removed for the authenticated professor. Publication itself does not create appointments or expose private events.

Implementation boundary

Reuse /appointments/availability and AppointmentService.publish_hours/remove_hours; preserve Instructor-only authorization.

USE CASE SPECIFICATION · UC-PROF-02 / CURRENT BACKEND

Review Student Appointment Requests

University Resource Allocation and Scheduling System

Primary actor: Selected required professor	Other actors: Student organizer; other required professors; optional invited students

Goal

Approve or decline a student-initiated appointment, with confirmation requiring every selected professor’s current-revision approval.

Precondition

The professor is a required participant on a visible Pending Approval appointment.

Main success scenario

1. Open My Appointments and the relevant request.
2. Review purpose, date, time, organizer, required professors, approval progress and invited-student status.
3. Choose Approve or Decline for the current revision.
4. On approval, the system rechecks required availability and records the decision.
5. If all required professors approve and the slot remains feasible, the appointment becomes Confirmed and reserves calendar time.

Alternative outcomes and constraints

- Decline: the request is Declined; no calendar reservation is created.
- Stale revision or new conflict: reject the outdated or impossible action.
- Organizer reschedules: all previous professor approvals are cleared.
- Organizer or required professor cancels: release the confirmed commitment and inform participants.
- Optional invited-student declines do not block professor confirmation.

Result

The correct decision, status, history, notifications and calendar consequences are persisted; pending requests do not reserve time.

Implementation boundary

Reuse /appointments/, /appointments/<id>, decision/cancel routes and AppointmentService; never turn optional invitations into approval vetoes.

USE CASE SPECIFICATION · UC-PROF-03 / CURRENT BACKEND

Manage Personal Schedule Blocks

University Resource Allocation and Scheduling System

Primary actor: Professor (Instructor)	Other actors: Calendar and scheduling services

Goal

Represent personal unavailability so valid appointment requests avoid blocked intervals.

Precondition

The professor is signed in and can access My Schedule.

Main success scenario

1. Open the personal schedule in month/week context.
2. Review authorized commitments and existing personal blocks.
3. Choose a start and end time, optionally adding a short reason.
4. Save the dated block; scheduling must account for the blocked interval.
5. Remove a block the professor owns when the interval is no longer unavailable.

Alternative outcomes and constraints

- Invalid or inverted interval: reject with an actionable message.
- Attempt to modify another user’s block: forbidden.
- Blocks cannot silently cancel or move already confirmed appointments or academic reservations.

Result

Only the professor’s own blocks change; subsequent availability checks consider them and calendar views stay consistent.

Implementation boundary

Reuse /appointments/calendar, AppointmentService.block/unblock, and PersonalScheduleService.

USE CASE SPECIFICATION · UC-PROF-04 / CURRENT BACKEND

Respond to Student Room Invitations

University Resource Allocation and Scheduling System

Primary actor: Invited professor (Instructor)	Other actors: Student organizer; authorized faculty room approver

Goal

Accept or decline optional participation in a student room activity without controlling the room reservation decision.

Precondition

A student submitted a room request that explicitly invited this professor.

Main success scenario

1. Open Room Requests and locate invitations visible to the professor.
2. Review the proposed activity, room, schedule and current reservation status.
3. Choose Accept or Decline for the current request revision.
4. If accepting, the system checks the professor’s relevant conflicts.
5. Display the professor’s response to authorized participants; the faculty approval lifecycle remains separate.

Alternative outcomes and constraints

- Conflicting professor schedule: the system refuses conflicting acceptance.
- Invitation or request revision is stale: refresh before responding.
- A declined invitation does not automatically reject or cancel the student room request.
- Only designated Staff/Administrator room approvers may approve or reject the reservation.

Result

The response is recorded with appropriate visibility and notifications; no new room authority is granted.

Implementation boundary

Reuse rooms invitation endpoints. Ordinary Instructor accounts are NOT faculty room approvers.

USE CASE SPECIFICATION · UC-PROF-05 / CURRENT BACKEND

View Calendar and Notifications

University Resource Allocation and Scheduling System

Primary actor: Professor (Instructor)	Other actors: University calendar; notification service; administrator publisher

Goal

See personally relevant appointments, invitations, dated commitments and explicitly published academic activities.

Precondition

The professor has an active authenticated account.

Main success scenario

1. Open the University Calendar and change month/week/day, date, or supported filters.
2. Review appointments and room invitations within the professor’s authorization scope.
3. View academic allocations only at the metadata visibility permitted by the publication policy.
4. Open Inbox to review appointment, room or calendar-related notifications.
5. Follow authorized links and mark notifications read.

Alternative outcomes and constraints

- Unpublished or private unrelated records remain invisible.
- An administrator’s optional instructor label is not proof of actual teaching assignment.
- Undated academic commitments require explicit administrative period binding; the UI must not invent dates.

Result

Professor sees a consistent, permission-scoped read model; calendar and inbox do not create a second scheduling authority.

Implementation boundary

Reuse /calendar/, /notifications, /appointments/, /rooms/. Academic publication and instructor attribution remain admin-controlled.

USE CASE SPECIFICATION · UC-PROF-06 / PROPOSED EXTENSION

Request Examination Room Allocation

University Resource Allocation and Scheduling System

Primary actor: Professor (Instructor) submitting own exam	Other actors: University administrator; allocation engine; academic calendar

Goal

Let a professor request exam space while the existing C++ strategies and authorized administration retain allocation authority.

Precondition

The professor is authenticated; the course and requested timing are valid under institutional policy. New professor submission authorization is not yet implemented.

Main success scenario

1. Choose an Exam academic request; supply course code/name, exam type, participant count and requested time.
2. Specify required facilities, building/space constraints and whether splitting across rooms is allowed.
3. Submit under the authenticated professor’s identity; the request is stored Pending, not allocated.
4. Authorized administration reviews the academic request and executes/publishes the existing C++ allocation workflow.
5. The professor sees their own request outcome, assigned room(s) or a clear rejection/conflict explanation.

Alternative outcomes and constraints

• Missing course metadata, invalid time/capacity or unauthorized faculty scope: reject.

• No feasible room or stale availability: request remains unconfirmed; disclose reasons and alternatives only if supported.

• If academic calendar periods are unresolved, show an explicit binding/configuration warning rather than inventing dates.

• Professor cannot select a final allocation, run a campus-wide strategy, or publish academic events.

Result

An instructor-owned exam request has an auditable Pending/Approved/Rejected outcome; only a successful authorized allocation reserves space.

Implementation boundary

NEW role-scoped intake/list/detail work is needed. Existing /add-request and /run-allocation are admin-only; C++ exam strategies already exist. Policy details require review.

USE CASE SPECIFICATION · UC-PROF-07 / PROPOSED EXTENSION

Request Teaching or Recurring Room Allocation

University Resource Allocation and Scheduling System

Primary actor: Professor (Instructor) submitting own teaching need	Other actors: University administrator; existing resource allocation engine

Goal

Request a suitable classroom or learning space for a lecture, lab, or recurring teaching activity without bypassing campus scheduling.

Precondition

Authenticated Instructor; permissible course/activity context and scheduling horizon. A professor-specific academic intake is not currently exposed.

Main success scenario

1. Choose OneTime or Recurring academic resource request.
2. Describe activity, attendance, time pattern, necessary facilities and space/building constraints.
3. Review validation, potential conflicts and the system’s actual supported input contract.
4. Submit the request as Pending under the professor’s own identity.
5. Authorized administration runs/publishes the existing resource-allocation workflow; professor reads outcome and relevant allocation details.

Alternative outcomes and constraints

- Insufficient capacity or incompatible resources: retain a clear non-confirmed status.
- Recurring patterns must use the existing supported weekday/time contract and academic calendar bindings.
- A preferred room does not imply guaranteed assignment.
- Academic requests must not be mixed with the student room-reservation approval workflow.

Result

A professor-owned academic request is tracked safely; confirmed resource allocation is authoritative only after authorized publication.

Implementation boundary

NEW professor submission and scoped result presentation; reuse existing OneTime/Recurring data contracts and C++ strategies without changing algorithms.

USE CASE SPECIFICATION · UC-PROF-08 / PROPOSED EXTENSION

Request Committee Meeting Space

University Resource Allocation and Scheduling System

Primary actor: Professor (Instructor) organizing academic committee	Other actors: Required committee participants; university administrator; allocation engine

Goal

Request meeting space and required attendance for an academic committee, thesis defense or project review using the existing committee-request domain.

Precondition

Authenticated Instructor; the university permits the activity; at least one active required committee participant is selected.

Main success scenario

1. Select CommitteeMeeting request and provide title, purpose, attendance, time and required resource.
2. Select the mandatory university participants and participant roles.
3. Validate participant identities, room/resource input and conflicts using the existing committee request contract.
4. Submit under the authenticated professor’s identity for administrative scheduling/allocation.
5. View the professor’s own outcome and permitted participant/resource details once processed.

Alternative outcomes and constraints

• No required participants, duplicate identifiers or invalid space/time: reject.

• Unavailable mandatory participant or resource: do not claim a confirmed meeting.

• A student-professor appointment is a separate domain; do not redirect this workflow through appointment approvals.

• Do not reveal unrelated users’ personal calendar details.

Result

An auditable committee space request is stored and may be allocated by the existing authorized academic workflow.

Implementation boundary

NEW Instructor intake/ownership/visibility. Existing CommitteeMeeting contract requires explicit space and mandatory participants; do not silently replace it with appointment logic.

USE CASE SPECIFICATION · UC-PROF-09 / PROPOSED EXTENSION

Track Own Academic Requests and Assignments

University Resource Allocation and Scheduling System

Primary actor: Professor (Instructor)	Other actors: University administrator; allocation result and calendar services

Goal

Give professors one read-only place to track their submitted exams/teaching/committee requests and explicitly attributed teaching events.

Precondition

A professor academic request or administrator-published instructor attribution exists; professor-specific query scope must be implemented.

Main success scenario

1. Open My Academic Requests to see only requests owned by the authenticated Instructor.
2. Filter by request type and state; open an authorized request detail.
3. See Pending, Approved, Rejected and allocation details backed by published engine decisions.
4. Show explicitly attributed published academic events in the professor’s calendar, without inferring assignments.
5. Follow activity-related notifications or calendar entries to the correct authorized detail.

Alternative outcomes and constraints

- A user cannot list or guess another professor’s private requests by ID.
- Unpublished or unbound academic results must not appear as confirmed dated commitments.
- An administrative instructor label is metadata, not proof of course ownership or permission to edit.

Result

Professors get accurate, scoped statuses and read-only assigned allocations without the ability to alter campus-wide scheduling.

Implementation boundary

NEW Instructor-scoped read models/pages; integrate existing allocation publication and university calendar projection. No new booking store.